

HL-EAI: A Multimodal Framework Enabling Emotional Reciprocity in Human–AI Strategic Decision-Making

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Abstract

This paper presents HL-EAI, a multimodal framework for studying emotion-driven cooperation in human–LLM, human-human, and LLM–LLM interactions via dynamic game-theoretic tasks. HL-EAI integrates emotional prompting, emotion recognition, and expressive AI avatars to enable bidirectional emotion transfer. By modeling affective influence on trust and alignment, it provides a testbed for developing emotionally aware agents. Our demo shows how multimodal emotional cues shape cooperation, advancing socially intelligent, human-compatible AI for interactive multimedia systems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Natural language processing**; **Computer vision**.

Keywords

LLM, Game Theory, Human–LLM Collaboration, Multimodal Agent

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1 Introduction

As large language models (LLMs) permeate interactive systems [4, 5, 7, 6, 19], their integration into multimedia environments demands attention to affective dynamics. Traditional alignment focuses on static benchmarks, overlooking emotion-driven trust and cooperation crucial in human–AI interaction [3, 27]. Yet, affective cues, expressed through language, vision, and audio, are central to effective collaboration [14, 17].

We introduce the HL-EAI framework^{1,2}, enabling controlled, multimodal experiments on emotional influence in human-human, human-LLM, and LLM-LLM decision-making in game theoretic settings. It supports interactive, prolonged strategy tasks where

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¹<https://github.com/mozikov/HL-EAI-framework>

²<https://youtu.be/vEJSkabB944>

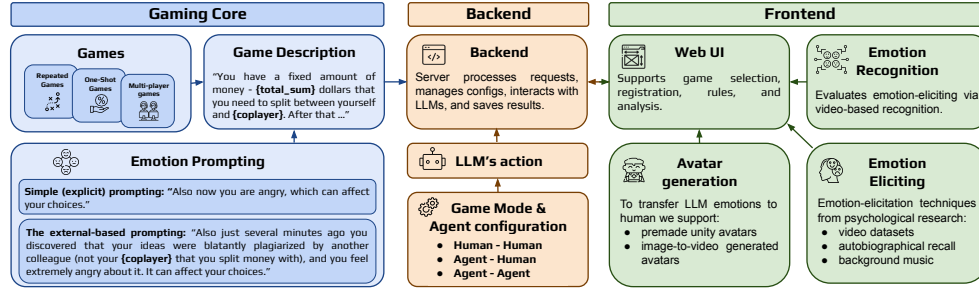


Figure 1: The HL-EAI framework uses game-theoretic scenarios to study emotional decision-making. Its backend manages configurations, while the frontend supports interaction, emotion handling, and visualization for human–LLM settings.

emotional cues shape outcomes, offering insights into trust calibration and alignment under dynamic conditions. HL-EAI lays the groundwork for next-generation multimedia systems where affective intelligence robustly enhances user engagement and trust.

2 Framework Overview

HL-EAI is a flexible framework for studying emotional and strategic alignment in human-human, human-LLM, and LLM-LLM interactions across diverse game-theoretic scenarios. Its core innovation lies in integrating emotional dynamics into decision-making, enabling nuanced analysis of trust, cooperation, and behavior. The platform supports one-shot and repeated games, including Prisoner’s Dilemma, Battle of the Sexes, and El Farol Bar, with extensibility through OpenSpiel (80+ games, basic coding required). HL-EAI features a modular architecture composed of a backend, frontend, and a gaming core with three components: **Game Description**, **Emotion Prompting**, and **Game-Specific Pipeline**.

2.1 Gaming Core

The **Game Description** module defines core rules and minimal context for three interaction types: one-shot (single decision), repeated (multi-round), and “ethical” (moral judgment) scenarios. LLMs are not assigned fixed personalities, allowing emotional prompting to be the primary experimental variable.

Emotion Prompting injects core emotions based on Ekman’s classification [2], following protocols from game theory [10, 26]. It supports three strategies: (1) context-free emotion priming, (2) co-player-directed emotion, and (3) external-context emotion, enabling analysis of how LLM behaviors align with known emotion-based human tendencies [12, 15]. After the initial prompt, ECOT [13] can be used for emotion-driven responses based on context.

The **Game-Specific Pipeline** applies these emotional states across one-shot bargaining games, repeated games with evolving strategies, and ethical dilemmas framed as single-shot decisions involving moral reasoning. Emotion and interaction histories evolve across rounds to simulate dynamic human-like behavior.

HL-EAI integrates various LLMs via APIs (e.g. OpenAI, Anthropic, OpenRouter), supporting models like GPT, Claude, and LLaMA. Configurable parameters (e.g., zero temperature, repeated trials) ensure consistent, verifiable outputs. This modular setup enables easy integration of new models and facilitates controlled emotional alignment experiments in diverse multi-agent settings.

2.2 Backend & Frontend

The backend manages configurations, real-time LLM interactions, logging, evaluation, and visualization. The frontend provides an intuitive web interface for both participants and administrators. It supports multilingual settings, real-time interaction, and visualizations to track decision-making behavior and emergent strategies.

To study emotional alignment, the frontend integrates multiple elicitation methods, including curated video datasets (FilmStim [24], EmoStim [25]), autobiographical recall [16], and emotion-inducing music [11]. With user consent, video-based emotion recognition [1, 9, 18, 22] detects participants’ affective states. HL-EAI supports the integration of AI avatars that embody LLMs and externalize their emotional states via facial expressions [8, 20, 21, 23, 28]. We support predesigned Unity avatars for consistent embodiment and video-based avatars for more realistic emotional transfer. Together, these components create a fully multimodal, emotionally responsive system for investigating human–LLM interaction dynamics.

3 Results

We used HL-EAI to study how emotions affect decision-making in human–LLM interactions during the Prisoner’s Dilemma. Thirty university students each played five games per emotional state on separate days. In anger–anger settings, both humans and GPT-4o showed a significantly sharper drop in cooperation than in the neutral case. In happiness–happiness settings, GPT-4o sustained high cooperation, and humans mostly reciprocated, with only minor declines. These findings suggest GPT-4o dynamically adapts to human behavior, and that emotional context modulates this adaptation. Anger accelerates mutual defection, while happiness promotes stability. This supports the idea that emotional alignment mechanisms can shape LLM behavior and that positive framing may help stabilize cooperation in mixed human–LLM environments.

4 Conclusion

HL-EAI demonstrates a multimodal framework for studying emotional alignment in human–LLM, human-human, and LLM–LLM interactions using game-theoretic tasks. By integrating emotion recognition, AI avatars, and dynamic emotional prompting, it enables rich, bidirectional emotional communication. Our results show that emotional context significantly shapes cooperation strategies, highlighting the potential of emotion-aware LLMs in socially adaptive AI.

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